Aakash Bansal

+1 (217) 298 6585 engage.aakash@gmail.com aakashban aakashbansalgit

Education

University of Illinois Urbana-Champaign **Aug 2024 – Dec 2025**
Master of Science in Financial Engineering GPA 3.72 / 4.00

Coursework: *Statistical Methods in Finance (A–), Machine Learning in Finance (A–) and Laboratory (A), Stochastic Calculus (A), Numerical Methods (A)*

Birla Institute of Technology and Science, Pilani **Aug 2018 – May 2022**
Bachelor of Engineering in Electronics and Communications, Minor in Finance CGPA 8.18 / 10.00

Coursework: *Mathematics I–III, Probability & Statistics, Principles of Economics, Development Economics*

Experience

Harvard Business School **Mar 2026 – Oct 2026**
Predoctoral Research Associate to Professor Rembrand Koning Boston, MA

- Ran a **Random Forest** on text embeddings that classified 47,007 startups as AI-native (AUC 0.81).
- Rebuilt the **replication package** for the resulting HBS paper on AI-native firms in R and Python.
- Designed a **field study** of heavy AI users with faculty, from research questions to the survey instrument.
- Audited the machine learning methodology and findings of a working paper in labour economics.
- Built Claude Code agents peers use from data to first draft, with checks against the source data.

Figure Lending **Jun 2025 – Dec 2025**
Capital Markets Intern, Data Science & Modelling Remote, New York City

- Built and deployed the **prepayment model** pricing Figure's HELOC securitizations, about $300M a deal.
- Estimated how borrowers respond to price, fitting the rate incentive that triggers refinancing per segment.
- Ran point-in-time backtests that rebuilt past deal pools and checked projections against outcomes.

J.P. Morgan Chase **Jan 2022 – Jul 2024**
Risk Analyst, Home Lending Credit Risk Bengaluru, India

- Built analysis panels from **400 GB** of Fannie Mae and Freddie Mac loan performance data.
- Wrote the **SQL logic** for delinquency buckets and status flags across a $200B mortgage book for CCAR.
- Built metro-level prepayment and default analytics across US housing markets.

Research Projects

Which News Predicts Market Volatility | *Python, GitHub* **Oct 2026**

- Tested which WSJ news categories predict volatility from 1926, excluding the text model's training years.
- Real-time forecasts beat a **HAR benchmark** from 1946 (Clark-West t = 4.0); all 35 specifications logged.

Machine Learning vs. HAR for the VIX | *Python, scikit-learn, GitHub* **Oct 2026**

- In walk-forward tests a month ahead, HAR beat a random forest (R^2 13.3% vs 6.5%, DM p = 0.003).

Skills

Languages: R, Python, Stata, MATLAB, SQL, SAS, Git, AWS

Econometrics: Panel regression, fixed effects, clustered standard errors, threshold regression, multiple testing

Machine learning: Random forests, gradient boosting, k-means, text embeddings, LLM APIs

Research practice: Replication packages, survey design, linking administrative data, specification logs